

Vermeers and Rembrandts in the Same Attic: Complementarity between Copyright and Trademark Leveraging Strategies in Software

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Abstract

We examine the interrelationships between firms' efforts to leverage different types of intellectual property (IP) rights. While prior research has tended to view different forms of intellectual property as substitutes, we argue that they can also be viewed as complements. We motivate this argument by shifting the unit of analysis away from the level of a single invention to the level of the technology, product or firm. Using unstructured interviews, we develop hypotheses about the causes of IP complementarity. Specifically, we hypothesize that complementarity may result from common inputs into the management of IP within the firm, such as managerial attention to IP issues and organizational resources deployed for IP management. Our hypotheses are tested in the PC software industry with a seemingly-unrelated-regression (SUR) model, which provides support for complementarity between copyrights and trademarks in software, and suggests that this complementarity stems at least in part from our hypothesized common inputs into IP management. Our work addresses both the relative neglect of non-patent intellectual property rights and the disproportionate reliance on the IP-as-substitutes view in prior research.

Keywords: Patents, Copyright, Trademark, Software, Litigation, Complements, Substitutes.

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INTRODUCTION

Firm strategy in the knowledge economy is increasingly aimed at generating rents from intangibles – at profiting from "Rembrandts in the attic." In response, management scholars have sought to understand how firms extract value from intangible assets, and have propelled the "appropriability mechanism" into our conception of firm strategy (Rumelt, 1984; Teece, 1986; Cohen et al., 2000). The use of intellectual property (IP) has been long recognized by managers and scholars alike as an important means of appropriating value, and recent management research has focused particularly on questions relating to different patent strategies pursued by firms (Somaya, 2003; Ziedonis, 2004; Graham and Mowery, 2004).

However, the almost exclusive attention given to patents in this literature has come at the expense of research on other types of IP. Because of this imbalance, research has tended to underplay the importance of non-patent forms of IP like trade secrecy, copyrights and trademarks, a view that does not square with reality. Patents have consistently been ranked among the least effective appropriability mechanisms in innovation surveys, especially in relation to secrecy (Cohen et al., 2000; Arundel and Kabla, 1998). We also know that litigation in copyrights and trademarks, which are perhaps the most neglected in scholarly studies of IP strategy, is at least as frequent as patent litigation in the U.S. (Somaya, 2004).

Earlier work has also largely ignored the *relationships* between different types of IP, in part because the field has concentrated so heavily on patents. Where these relationships have been implicitly considered in the literature, the underlying assumption has generally been that different types of IP are substitutes. This view is borrowed from economics and law, where it is based primarily on the relationship between patents and trade secrets (Friedman et al., 1991; Horstmann, et al., 1985; Teece, 1986). Because patenting requires information disclosure, which necessarily vitiates secrecy, a substitution effect between different types of IP seems obvious. However, the possibility that different types of IP may act as complements, and if so under what circumstances, are questions that could fundamentally alter our conception of IP strategy, and therefore deserve greater attention.

Management scholars have long appreciated that individual appropriability mechanisms can interact with other firm assets or market dynamics, with powerful performance implications. Teece (1986) for example explored the relationship between appropriability and complementary assets, while Connor and Rumelt (1991) examined the tradeoffs between asserting IP rights against piracy and capitalizing on positive network externalities. But such systematic analyses of interactions have not been extended to different types of IP protection beyond the patent-secrecy tradeoff. In part, this omission is due to difficulties in seeing beyond the IP-as-substitutes view so long as one maintains the invention as the unit of analysis. In this paper, we argue that by shifting the unit of analysis to an aggregate level such as the product or the firm, the potential for recognizing the complementarities between different types of IP is unleashed. Hence firms may simultaneously leverage different types of intellectual property together to appropriate greater value from their knowledge assets, rather like finding Rembrandts and Vermeers together in the same attic.

The relative paucity of non-patent IP research may have its genesis in the non-availability of data, especially in contrast with the ubiquitous and rich data on patents (Griliches, 1990). Our paper provides a workable solution to this “data problem” by developing litigation data to examine trademark and copyright use by firms. While prior work has studied a range of appropriability mechanisms through “innovation surveys” (Levin, et al., 1987; Harabi, 1994; Arundel and Kabla, 1998; Cohen, et al., 2000), secondary litigation data has the advantage that it measures actual firm behavior in relation to IP. In addition, as we discuss below, firm-level copyright and trademark litigation data do not suffer from many of the limitations of copyright and trademark registrations data, which have made them less useful for research (Graham and Mowery, 2003).

Our research is set in the software industry, where the variety of intellectual property protection used by inventors – patents, trade secret, copyrights, and trademarks – offers a uniquely promising setting to study complementarity and substitution in IP. Furthermore, the sudden legitimization of software patent protection in the early 1990s acted as an exogenous shock to the industry, spurring different firms to react in idiosyncratic ways to the changed circumstances. Our empirical methodology exploits this

variation, allowing us to examine if the resulting shifts in firm behavior provide evidence for complementarity or substitution in intellectual property strategies.

COMPLEMENTARITY AND SUBSTITUTION IN INTELLECTUAL PROPERTY

The view of intellectual properties as substitutes is writ large in management research. Because intellectual property is a creature of the law, and because economists began analyzing IP protection quite early (Plant, 1934a,b), it is not surprising that these disciplines had a strong influence on our understanding of the relationships between different types of IP. In the law, the substitutes view has been driven by the strength of precedent: in *Kewanee Oil versus Bicron Corp.* (416 U.S. 470 (1974)), the U.S. Supreme Court declared that trade secrets were appropriate for “lesser or different inventions” than patents. In their economic analysis, Friedman et al. (1991) disagreed with the Court's opinion, but nevertheless maintained the assumption that patents and trade secrets were substitutes.

An important contributor to the IP-as-substitutes view in research has been the dominant focus on patents, and researchers' routine practice of evaluating tradeoffs between patents and trade secrecy. The premise that patent rights and trade secret protection are natural substitutes has an influential pedigree in theoretical studies of appropriability (Kahn, 1962; Machlup, 1962; Horstmann, et al., 1985; Teece, 1986). Both Kahn (1962) and Machlup (1962) propose that inventors choose trade secret as a substitute for patent protection to control information disclosure. Formalizing this logic, Horstmann et al. (1985) show that if firm profits are expected to fall due to patent disclosure, there is a reduction in the equilibrium likelihood of patenting, an outcome they suggest is the substitution of trade secrets for patents by the firm. Teece (1986), a fountainhead for much of the recent research on appropriability strategy, also reinforces the IP-as-substitutes view by pointing explicitly to trade secrets as an "alternative to patents" (p. 287).

Consistent with these theoretical treatments, empirical findings have also persuaded management scholars to think about different types of IP as substitutes. In the Yale study (Levin et al., 1987), the authors use correlations and factor analyses to reduce appropriability mechanisms to two dimensions, patent and non-patent (including secrecy), which appear to be used as alternatives. Harabi (1995) surveyed Swiss firms to evaluate the relative importance of patenting and secrecy, assuming implicitly

that firms must choose between them. Using a Europe-wide survey, Arundel and Kabla (1998) found a negative correlation between the perceived effectiveness of trade secrecy and (product, but not process) patenting rates, a finding that is consistent with a substitutes view. Further, Arundel (2001) used data from the same survey to compare the use of patents and trade secrets in R&D-performing firms, under the assumption that these are substitutes.

Fundamentally, the IP-as-substitutes view can be traced to the prevailing practice of adopting the “invention” as the unit of analysis. When focusing upon a single invention in isolation, the use of one type of intellectual property is likely to decrease the marginal benefit obtained from using a second type. For example, prior to 1992 it was thought that non-literal elements of software programs (such as menu organization) were adequately protected by copyright, and therefore additional patent (or trademark) protection offered relatively small benefits. Moreover, legal or practical considerations may preclude the use of some types of IP when the invention is already protected by others. The quintessential case occurs between patent and trade secrecy because the patent law requires the invention's disclosure, making it impractical (and possibly illegal) to simultaneously protect the invention as a trade secret (35 USC 112 (2005)). The law also buttresses the IP-as-substitutes view for individual inventions, axiomatically assigning patents to protect only ideas, and copyrights to protect only expression (*Baker v. Selden*, 101 U.S. 99 (1879)).

The Case for IP Complementarity

While there is considerable support for the IP-as-substitutes view when we focus on *a single invention at a fixed point in time*, a more nuanced approach may be appropriate when one moves away from this unit of analysis. Fundamentally, such a move requires us to recognize that any technology or product may comprise numerous *separately protectable* technological elements (i.e. inventions). For example, HiddenMind Technology, Inc., a mobile-workforce software company, protected its ActiveUniverse® brand name by registering the trademark (SN: 75907093 in 2000), obtained a patent on the associated product idea (PN: 6442565 granted in 2002), and also enjoyed copyright protection for the software source-code. This example is by no means unusual. In prior work, Arora (1995) has highlighted

how German chemical firms used patents *and* trade secrets to protect *different* elements in the firms' value chains. However, this study did not explore interactions between the firms' use of patents and trade secrecy, adopting a view that trade secrecy was more appropriate for tacit manufacturing know-how and patents for codified product inventions (a treatment suggestive of the “substitutes” view, but one which was adequate for the questions about licensing that were the focus of the study).

When examining intellectual property strategy broadly at the level of technologies, products, or even firms, IP complementarity is no longer limited by a choice between mutually exclusive appropriability alternatives. For example, while firms may be constrained to use either trade secrecy or patent to protect the *same* invention, they may use both simultaneously for separate related inventions. Such opportunities are made even more explicit in the case of IP rights that cover different types of “inventions,” such as patents and trademarks. For example, Teece (1986) describes how Searle (since acquired by Monsanto) protected its artificial sweetener Nutrasweet® both with patents (for product inventions) and with trademarks (for the brand name), and appropriated greater value as a result. Moreover, the value generated (subsequently) by the firm's trademark use was a result, at least in part, of its successful use of patents to build up the Nutrasweet franchise.

The Nutrasweet example conforms to the classical definition of complementarity in firm strategies – increased use of one strategy (leveraging trademarks) generates greater benefits for the firm when another strategy (leveraging patents) is also employed (Milgrom and Roberts, 1990). Formally, two types of IP can be said to be complementary when the cross partial of firm value “V” with respect to the use of both types of IP is positive (Arora, 1996; Athey and Stern, 1998), i.e. $\left[\frac{d^2V}{d(IP_1)d(IP_2)} > 0 \right]$. By aggregating up our unit of analysis from the level of a single invention to the set of inventions in a particular technology, product, or firm, we demonstrate the feasibility of an IP-as-complements view. At the heart of this view is the idea that the leveraging of one type of IP for some inventions may generate higher firm value when another type of IP is also simultaneously leveraged for other related inventions.

Empirical studies also lend some support to an IP-as-complements view. In addition to the work by Arora (1995) discussed above, factor analyses done on the Carnegie-Mellon appropriability survey showed that several appropriability mechanisms often load together, suggesting complementarity at the level of the firm (Cohen et al., 2000). Specifically, product and process patenting are shown to load on the same factor with "other legal," and in some cases with secrecy. Although the survey does not clarify whether "other legal" reflects legal intellectual property (such as copyright and trademark) or other areas of law (such as anti-trust and contract), the results cast some doubt on the IP-as-substitutes view.

To further understand the relationship between different types of intellectual property, and to inform our theory development, we conducted a series of interviews with experts in the field. We restricted ourselves to managers in firms included in our dataset, and contacted 25 executives identified as IP legal counsel or IP managers in 19 of the top 100 U.S. packaged software firms by revenue (2001). Only 16 of these executives were still employed at their respective firms, and 6 individuals (38%) at 5 firms agreed to be interviewed. These firms each had greater than \$50 million annual revenues and 500 employees in 2001. Using open-ended unstructured interviews we asked each respondent whether software firms were likely to use different types of IP together in a complementary manner. We further investigated – to the extent that different types of IP were being used together – what factors might be driving the complementarity in IP. Our interview results are summarized in Table 1.

[Table 1 about here]

The responses to our questions were broadly supportive of complementarity in the use of different types of IP in software. Our respondents generally agreed with the premise that different types of IP were used together, using words like “it seems natural” (manager 4) or “it’s not surprising to me” (manager 5). Only one respondent (manager 3) from a firm known to be unfriendly to software patenting expressed some reservations (about allowing software patents), but seemed to agree that other companies behaved in this way because they “have created profit centers ... to monetize their IP.” One IP manager (manager 2) even described an evolution in thinking about IP at his/her firm, where different types of IP were initially viewed as separate, but are now managed in an integrated way.

These responses, and our deductive reasoning above, provide support for the possibility of complementarities between different types of intellectual property at the level of a product or a firm. We recognize however that IP substitution may be common at the level of a single invention, and that these invention-level effects could aggregate up to the level of the product, technology or firm. It may be that an increasing focus on using one type of IP to protect a product, or within a firm, may instead decrease the value provided by other types of IP. Therefore, while we argue that complementarity between different types of IP is feasible, we remain agnostic as to whether or not that will be the case. We therefore frame the following alternate hypotheses.

H1: Intellectual property rights may be used as substitutes when examined at the level of the product or the firm.

H1A: Intellectual property rights may be used as complements when examined at the level of the product or the firm.

Drivers of IP Complementarity

Our approach to complementarity in intellectual property is inherently exploratory, and is driven by the basic intuition that when we broaden our unit of analysis from the level of an individual invention, opportunities for using different types of IP together in a complementary manner become available. Complementarity in firm strategies can be categorized into two broad types based on the mechanism through which the complementarities are generated – market-driven complementarity and production-cost complementarity. Market-driven complementarity is essentially caused by demand conditions, wherein the increased use of one type of strategy alters market conditions so as to increase the marginal value of also using another. For example, more aggressive patent protection may prevent the sale of close substitutes, thus helping a firm to raise product prices. Such a lucrative product is likely to attract copycat products sold illegally under the same brand name, which would in turn increase the value of a more aggressive trademark strategy from the firm. Lybecker (2003) uses a similar argument to conclude that stronger patent rights for pharmaceuticals are likely to elicit counterfeiting, and therefore require stronger anti-counterfeiting enforcement.

Another type of complementarity can arise when production costs decrease because interrelated strategies are deployed together, such as the cost savings generated by modern manufacturing strategies – flexible equipment, fast order processing, low inventories, fewer job categories – when used jointly in the firm (Milgrom and Roberts, 1990; Milgrom, Qian, and Roberts, 1991). While such production-related complementarities may be driven by the inherent interactions between firm strategies, they may also stem from the use of common indivisible “factors of production” across multiple activities, which give rise to a form of scope economies (Chandler and Hikino, 1990). For example, if an input used for leveraging one type of IP can also be employed to leverage another type of IP at relatively low additional cost, it may be more valuable for the firm to leverage both types of IP together, rather than each one separately – which by definition implies complementarity between these types of IP. By and large, our interviews provide strong support for this source of complementary in software firms' use of IP. Accordingly, and because we lack a means of directly testing for market-driven complementarity, we focus the balance of our discussion on key types of common inputs that are critical for effective IP strategy and management. We identified from the literature two such inputs that appeared especially relevant to IP strategy, which were also strongly highlighted by our interview respondents: (1) *Management attention to IP* and (2) *IP-related organizational resources*. One of the IP Managers we interviewed went so far as to identify both of these specifically as the drivers of firm behavior (manager 3).

Managerial Attention to IP

Herbert Simon (1947) first examined the implications of limited human attention capabilities for organizations and their administrators. Simon saw human attention as a scarce resource, one that contributes significantly to bounded rationality because individuals do not fully incorporate all possible alternatives or consequences when making decisions. Building on the work of Simon and subsequent scholars (Cyert and March, 1963; Weick, 1979; Cohen et al., 1972), Occasio (1997) formulated an attention-based view of the firm, the central argument of which is that firm behavior can be substantially explained by the issues and answers focused on by its key decision makers.

Managerial attention can therefore be thought of as a scarce resource that is an input into firm decision-making, and thus into value-adding actions taken by the firm. Because attention-generating processes can be idiosyncratic and context-specific (Occasio, 1997), they may be difficult for other firms to reproduce, and are therefore a potential source of sustainable rents for the firm (Peteraf, 1993). Moreover, as firm management begins to pay greater attention to a given issue – for example, IP management – managers also begin to acquire greater awareness about its potential ramifications. Accordingly, once managerial attention is focused on IP-related issues and awareness about IP is developed, firms may find it cost-effective (especially in relation to information costs) to leverage multiple forms of legal IP, where previously these potentially value adding strategies may have been ignored by the firm. This is an illustration of classic common-input scope economies that can lead to complementarity in IP strategies. Inherently, this explanation assumes an evolutionary process, whereby firms discover sources of value by imperfect search and discovery (March, 1991). But, having focused their attention on IP, firms are able to uncover opportunities to profit from on multiple related IP assets through low-cost “local” learning.

We find considerable support in our interview responses for the role of managerial attention in generating IP complementarity. Several respondents discussed the evolution of IP management at their firm, whereby increasing attention to IP and IP strategies in the firm gave rise to “thinking about these issues holistically” (manager 6). Other managers described this process as one where “the organization and the management has [sic] become more sophisticated about [IP] issues,” (manager 2) and one that “sensitize[d] business managers inside the company on [sic] IP issues” (manager 1). In each case, the increased attention to IP strategy among firm managers was presumed to result in the use of different types of IP in a complementary manner. As one IP manager put it, “if a ... company begins using trademark, it is much easier to begin using patent and copyright” (manager 1).

H2: The use of intellectual property rights as complements may be driven by greater attention to intellectual property by firm management.

IP-Related Organizational Resources

A second factor highlighted by our interviewees is the development of firm-level expertise in the area of IP law and IP management, which was reported by them to be an important organizational resource for conducting IP strategies. This factor was described variously as the hiring of “in-house counsel with expertise in IP” (manager 1), the “increasing sophistication of [the company’s] IP legal department” (manager 2), or even the “legal department theory” (manager 3). One contribution of marshalling these organizational resources appears to have been to “sensitize” or “evolve” managerial attention towards IP-related issues and strategies, as discussed above. In addition, the ability of IP experts to manage and leverage IP by virtue of their own domain knowledge and by educating firm management was also recognized (managers 1 and 2). IP expertise may therefore be viewed as a somewhat sticky (or specialized) organizational resource that can generate value for the firm in conjunction with its associated unique invention assets (Lippman and Rumelt, 2003).

Because IP attorneys tend to have broad training and expertise that spans the IP domain, this organizational resource may be leveraged across multiple types of IP. For example, the same attorneys who identify opportunities for adding to the firm’s patent portfolio may also design copyright or trademark strategies to fight piracy and counterfeiting.¹ As our respondents put it, “[l]awyers throw all the tools in the box at you” (manager 4), and “I would naturally expect that if a company is trading in one type of IP it will value all types ... [l]awyers will use what they have on hand: if a tool is available, the tool will be used by the company” (manager 2). Some elements of IP expertise may be obtained by using outside counsel (manager 1), but there are contractual impediments in transacting for a service that is necessarily company-specific and also highly strategic (Williamson, 1985). Moreover, even if some IP needs are outsourced, internal expertise may be needed to knowledgeably manage the outsourcing

¹ Our consultations with experts produced a mixed picture of the extent to which IP attorneys are “generalists,” and therefore able to undertake activities relating to different types of IP. Broadly, at large firms with heavy IP workloads there is some specialization, especially at lower levels, while attorneys at smaller firms and at higher levels tend to be generalists whose knowledge and work span across IP types.

relationship. Accordingly, we conclude that the organizational resources accumulated by firms to manage and leverage IP may have played a significant role in the complementary use of different types of IP.

H3: The use of intellectual property rights as complements may be driven by organizational resources that the firm can employ across different types of IP.

SOFTWARE AND INTELLECTUAL PROPERTY

The packaged software industry in the U.S. had grown to nearly \$70 billion in revenues by 2001.² During the 1980s, swift adoption of the desktop personal computer (PC) created a fast-growing need for specialized PC software, which was met mostly by specialized market entrants like Microsoft and Adobe Systems. Software is somewhat unique in that all four major types of intellectual property – patents, copyrights, trade secrets, and trademarks – are commonly used for appropriability in this industry. While trade secrets are protected against misappropriation by state laws, federal laws provide the property rights entailed in patents, copyrights, and trademarks, these latter being the focus of our analyses.

Federal trademark protection is primarily used in this industry to protect identifiable logos and brand names associated with a software company or its product. Trademarks protect some of the most valuable assets of software companies – for example, the Microsoft brand name alone was valued at \$59.9 Billion in 2005 (Berner and Kiley, 2005). Software firms use copyrights primarily to protect the written code in their programs from literal copying. While copyright protection was extended to software by the U.S. Congress in 1980 (Samuelson, 1984), software's patentability is an even more recent development in the U.S., established primarily by changes in patent office practices and case law in the courts. While there is debate about when the trend actually began (see Graham and Mowery, 2003), the creeping institutionalization of software patenting in the 1980s was solidly in place by the early 1990s (Department of Commerce, 1992). The early 1990s were more generally a key turning point in the legal treatment and firms' use of software intellectual property. Along with the increased acceptance of patents, copyright

² US Census Bureau (online: http://www.siia.net/software/pubs/growth_software05.pdf, accessed 2/23/2006).

protection was also curbed by the courts, thereby increasing the *relative* rewards to patenting.³ The changing intellectual property environment for software in the 1990s was reflected in the rapid growth of software patenting and patenting per R&D dollar, even in comparison to the overall growth rate of patenting across industries in the U.S. (Graham and Mowery, 2003).

DATA AND METHODS

Data and Sample

Defining the boundaries of “the software industry” is a notoriously thorny challenge because software is a general-purpose technology used in a wide range of applications. We focused our attention on the (IBM) PC packaged software industry, and included all top-100 firms in this segment from 1985 to 1999 (by revenue, obtained from a commercial firm, Softletter). This industry definition has the advantage that PC software firms tend to be software specialists, which enables us to identify “software IP” by observing the identity of the firm. During this period, the majority of these firms were highly focused on the PC market – Softletter’s criterion is that firms on its list generate at least 50% of sales from PC software. Our sample of firms contains virtually all the significant firms in this industry – for instance, the smallest firm in 1999 had revenues of only \$ 4 million.

We identified all copyright, patent, and trademark litigation involving the firms in our sample by matching on firm names, using IP litigation data from the Federal Judicial Center (FJC).⁴ Since the data on firms’ uses of all types of IP were obtained from the same source and identified using the same techniques, biases stemming from idiosyncrasies of different data sources or sampling techniques are minimized. Mergers and acquisitions among firms were also tracked to ensure that ongoing litigation was assigned to the correct successor firms.

³ *Computer Associates Int’l v. Altai* (982 F.2d 693 (2nd Cir. 1992)), and *Lotus Development v. Borland International* (49 F.3d 807 (1st Cir. 1995), aff’d 116 S.Ct. 804 (1996)).

⁴ Federal Court Cases Integrated Database (various versions, 1970-2001), Ann Arbor, MI: Inter-university Consortium for Political and Social Research (distributor). IP litigation was identified using firm-name search strings constructed to err on the side of inclusion (85% of suits were subsequently discarded). Manual checking confirmed that matches were accurate, with thorough research in secondary sources (news articles, company information) when needed. If the identity of the firm could not be confirmed, the suit was excluded (< 3% of cases).

Figure 1 charts the trends in patenting and patent litigation 1985-1999. Consistent with the changes affecting software IP, both patenting and patent litigation experienced a many-fold increase in the 1990s. However, the software firms in our sample are defendants, not plaintiffs, in the majority of these suits. Further, Figure 2 illustrates that the increase in copyright litigation is no less dramatic than patent litigation, and begins from a much higher base (48 copyright suits in 1990 as against 7 patent suits). Similarly, there has also been a substantial rise in trademark litigation for PC software firms in the 1990s, again starting from a somewhat higher base. Thus, while widespread patenting has been heralded as an important shift in the software industry, the growth in use of copyrights and trademarks has been no less remarkable, at least in the PC packaged software industry.

[Figure 1 and Figure 2 about here]

To examine our central question, whether there is complementarity between firms' use of different types of IP, we focused on copyrights and trademarks, employing litigation data to measure the firms' "use" of these protections.⁵ Litigation data have significant advantages over data derived from copyright and trademark registrations for studying IP use by firms. Copyrights accrue to the inventor by the mere act of "fixation" (writing the software code), and copyright registration is a relatively insignificant act, except in the event of litigation.⁶ The direct costs of registration are also very low (a two-page application form and a filing fee of \$30), and there is no examination of the registration itself (thus a software program may be registered as a single or as multiple copyrights). Further, because there is no compelling incentive to register copyrights early, there is often a random bunching of registrations over time. For these reasons, copyright registration data tend to be extremely noisy (Graham and Mowery, 2003). Trademark registration is also relatively low cost (about \$1000, including legal costs),

⁵ We exclude patent litigation data in our analysis because, unlike copyrights and trademarks, the majority of patent suits in our data were filed *against* PC software firms, which cannot reasonably be described as a *use* of patent rights by these firms. Once these suits (and retaliatory actions) are dropped, we are left with only a very small number of cases initiated by our sample firms. However, as noted in the text below, patents and in particular the increase in patenting in the 1990s, play an important role in our empirical findings.

⁶ Copyright infringement suits cannot be filed in U.S. courts until the copyright is registered. Plaintiffs are denied recovery of attorney fees and statutory damages for infringers' acts conducted prior to the date of registration, but these damages are generally small compared with actual damages. Thus, registration is necessary as a prelude to litigation, but not otherwise. 17 U.S.C.A. Sections 411, 412, 504, 505 (2000).

but while an examination is required to evaluate whether the mark is valid, trademark registration data suffers from many of the same drawbacks as copyright. Trademarks are likely to be registered as a matter of course, and registrations may merely reflect the creation of new marks by the company rather than any concerted effort to capitalize on the protection offered by the property right. By contrast, litigation data provide a useful alternative measure of IP use, because the costs of trademark and copyright litigation are non-trivial, and litigation clearly reflects a choice to leverage the firm's property rights for strategic gain.

In addition to IP litigation data, we collected copyright count data from the Library of Congress, patent applications, trademark filings, and a count of registered patent attorneys employed by each firm from the United States Patent and Trademark Office (USPTO), along with public accounting data for our firms from Research Insight®. The Softletter-100 lists also provide data on founding years, annual revenues and employment for each firm, which supplement the Research Insight® data. After accounting for missing data, we are left with an unbalanced sample of 88 software firms over the time period 1985-1999. Since many of the changes in software IP use occurred in the early-to-mid 1990s, and we employ panel data statistical techniques, this time frame is an ideal one with which to study our research question.

Statistical Methods

We evaluate substitution or complementarity between copyrights and trademarks by examining the extent to which firm decisions to leverage both types of IP “move” together. Ordinary cross-sectional regressions of one type of IP on another are inappropriate to this task. The estimates from such a model would be subject to omitted variable biases (Arora, 1996; Athey and Stern, 1998); that is, a firm's use of both types of IP may be correlated with other unobserved variables, thus producing a spurious complementarity (or substitution) finding. To overcome this problem, we model the use of each type of IP separately using panel data techniques, and account for alternative explanations of IP use in each separate regression equation. To test for complementarity (or substitution), we stack these separate IP equations together and examine the residual correlation, or non-independence, between the unexplained errors. If these residuals are positively correlated, it suggests the existence of complementarity, whereas

negative correlation suggests substitution. This is essentially Zellner’s seemingly unrelated regression (SUR) model (Greene, 1997, pp. 674-703), where copyright (C) and trademark (T) use are modeled (in vector notation) as follows.

$$\begin{aligned} C &= X_C b_C + e_C \\ T &= X_T b_T + e_T \end{aligned}$$

We include firm fixed effects and year dummies to control for idiosyncratic differences between firms and years, and to mitigate omitted variable bias. Thus, we are accounting for inter-firm differences, and focusing exclusively on correlations between within-firm changes in IP use. Finally, we introduce each of our key independent variables (derived from hypotheses H2 and H3) to examine if they explain the observed residual correlation between the use of copyrights and trademarks within each firm. This approach is appropriate for understanding the nature and drivers of IP complementarity, which is our primary research interest. To avoid concerns about endogeneity and reverse causation we lag all the independent variables by one time period (year). Finally, we address left truncation at zero in our dependent variable by using a dummy variable for all such observations, a technique that has been employed previously to address similar concerns (Pakes and Griliches, 1980).

Variables and Measures

Dependent Variables.

In our two-equation model, the two dependent variables are the use by firms of copyright and trademark, the measure for each being derived from litigation data. However, merely counting the suits filed in any given year (i) weights all IP suits equally, irrespective of their significance, and (ii) places all the weight of litigation on the year in which the suit was filed rather than distributing it over the life of the suit. We address these measurement issues by using the total suit-years of IP litigation in any focal year as the measure of IP use by a firm. We count the number of days for which each suit’s docket was open in a given year, and sum these suit-days across all suits of the same type – copyright or trademark – for each firm. Finally we convert these days to years to obtain our suit-year dependent variables. For

example, a firm i engaged in two copyright suits lasting the entire year t would produce a measure of 2.0 copyright suit-years for that firm-year observation (C_{it}).

Independent and Control Variables.

As described above, we measure *substitution or complementarity* (H1 and H1A) in firms' IP use by examining the correlation between the unexplained residuals in our model. We measure *managerial attention to IP* (H2) by using the level of patenting activity within each firm. From Figure 1 we can see that software firms significantly increased their patenting during the 1990s, possibly in response to the strengthening of software patents discussed earlier. Our interviews indicate that this move into patenting had the effect of sensitizing firm management to IP-related issues more broadly, and led them to focus on IP as a whole in a strategic manner. In software particularly, where engineers were widely opposed to patenting on philosophical grounds (Samuelson and Glushko, 1990), pursuing patents was generally a choice made by firm management, and therefore reflects managerial attention to IP strategies.

Our measure for *organizational IP resources* (H3) is the number of patent attorneys employed by the software firm. We identified these by using the roster of patent attorneys maintained by the USPTO's Office of Enrollment and Discipline (OED). To become a patent attorney, a lawyer must have technology training (usually at least a bachelor of science) and pass a competitive patent bar examination. Therefore, patent attorneys are a well-defined group of highly qualified professionals, and we are confident in using the patent office's rosters to identify these workers and their employers because the OED requires that attorneys maintain their current address on file, and periodically conducts audits to ensure accuracy. Failure to comply can result in a revocation of the attorney's license to practice at the USPTO.

In addition to these key independent variables, we also used a number of control variables that may otherwise account for changes in the use of copyrights and trademarks by a firm. These include firm revenues, firm age, number of employees, total assets, R&D expenditures, and pre-tax profits. We also employ a count of copyright and trademark registrations in the copyright-use and trademark-use equations, respectively. As discussed earlier, we also include a dummy variable for observations left-censored at zero. All financial data are converted to 1990 dollars using a GDP deflator.

RESULTS

The descriptive statistics of our data are reported in Table 2. We exclude the correlation table because our estimation is based on within-firm variation, whereas raw correlations are swamped by between-firm variation and are therefore not meaningful. The SUR model was estimated using Generalized Least Squares (GLS), the results of which are reported in Table 3. Each run of the model is reported in a panel of two columns, corresponding to the copyright and trademark equations respectively. The first pair of columns in Table 3 reports results from independent fixed-effects regression models for the copyright and trademark equations separately (Model(s) 1). Hausman's test for independence of the firm effects from the other variables is easily rejected, indicating that a fixed effects model, and not random effects, is appropriate for this sample. F-tests for the model as a whole are significant in all models reported in this paper.

[Table 2 and Table 3 about here]

Model 2 employs the same variables, but estimates the SUR – that is, it allows for correlation between the residuals. The estimated residual correlation in this model is positive, large (+ 0.31), and strongly significant ($p < 0.01$). This result demonstrates that after accounting for mean firm-level effects and control variables, higher (lower) within-firm levels of copyright use are associated with higher (lower) within-firm levels of trademark use. This positive correlation provides prima facie evidence for complementarity between software firms' copyright and trademark use (H1A is supported). We find no empirical support for IP substitution at the level of the firm (H1 not supported).

Model 3 estimates the SUR with the addition of our *managerial attention* variable (H2). The coefficient of this variable is large and highly significant in both the copyright and trademark equations. Each additional patent filed by the firm is associated with 41.2 additional suit-days of copyright litigation and 6.9 suit-days of trademark litigation (each significant at the 0.01 level). At the same time, the correlation between the residuals in the SUR fell to 0.25, a large and statistically significant reduction ($p < 0.05$; estimated by simulation). Thus, there appears to be strong evidence that firms which increased their managerial attention to IP strategy (proxied by patent use) also experienced significant increases in

copyright and trademark use, and that this effect accounts for a significant fraction of the observed complementarity between the two types of IP (H2).

In Model 4 we omit the managerial attention variable while including our *organizational IP resources* variable. The coefficient of this explanatory variable is also large and highly significant in both equations. An increase of one patent attorney in the firm is associated with 729 additional suit-days of copyright litigation and 168 suit-days of trademark litigation (again significant at the 0.01 level). As in the last model tested, there is a significant drop in the correlation between the residuals in the SUR (to 0.25). This result suggests that organizational resources used for IP management may also be a mediating mechanism that explains complementarity between copyright and trademark use in software (H3).

When we evaluate the full model (Model 5), we find that organizational IP resources do not have an independent effect in explaining IP complementarity. Indeed, relative to Model 3, we can even reject the hypothesis that the addition of this variable significantly increases the explanatory power of the model. In other words, it appears from these results that the effect of organizational IP resources is strongly correlated with the effect of managerial attention to IP. This finding is not entirely surprising, as our interviews suggest that a significant role of organizational IP resources was to focus the firm's attention on IP.

The Microsoft Effect

In our sample, one firm is unique because it dominates the PC software industry and exerts substantial control over PC standards. Microsoft's revenues in 1999 were 16 times larger than the next largest firm, and its profits were 20 times greater than the next most profitable firm. Naturally, we were concerned that Microsoft may be an outlier in our data, and that its IP behavior may also be idiosyncratic. We therefore re-estimated our models after removing Microsoft from our sample (Table 4).

[Table 4 about here]

Dropping Microsoft from the sample has some effect on the magnitudes of our estimates, but does not fundamentally change their character. Upon rerunning the base SUR model (Model 2), we still find a positive and statistically significant correlation between the residuals (+ 0.26). Similarly, when we

add our independent variables, *managerial attention to IP* and *organizational IP resources*, there is a statistically significant drop in this correlation ($p < 0.05$), and the coefficients of these variables are positive and significant (Models 3 and 4).

Strikingly, organizational IP resources appear to play a more prominent role in explaining IP complementarity when Microsoft is no longer in the sample. When we rerun the full model (Model 5), the *organizational IP resources* variable is statistically significant in each equation ($p < 0.05$) and it significantly increases the explanatory power of the model as a whole. Moreover, there is a significant decrease in the correlations between the residuals ($p < 0.05$), our measure of complementarity, when this variable is added (to Model 3). This result suggests that with "smaller" firms (other than Microsoft) organizational resources may explain IP complementarity over and above managerial attention alone. And, it is consistent with the view that in smaller firms, IP managers are less specialized in their work and therefore their expertise is likely to be leveraged across multiple types of IP, which may partly explain the observed complementarity. In the full model (Model 5), without Microsoft in the sample, we find that the residual correlation between the errors falls essentially to zero. It thus appears that all of the complementarity we observe between the leveraging of copyrights and trademarks in our sample can be explained by three factors – managerial attention to IP, organizational IP resources, and a Microsoft-specific complementarity effect.

As an additional sensitivity analysis, we also estimated the SUR model by dropping the top 25% of firms (by revenues). In this sample, we continued to find a statistically significant correlation between the unexplained residuals, suggesting IP complementarity (H1A), but the other independent variables had no statistically significant effect. This finding is consistent with other industry studies in which support for the theory is weaker among the smaller firms in the sample. Moreover, since over 95% of patents and virtually all patent attorneys come from the top quarter of PC software firms, it is not surprising that we were unable to find support for H2 and H3 in the rest of the sample.

DISCUSSION AND CONCLUSION

This paper is among the first to focus attention on the strategic use of non-patent forms of intellectual property. Our finding that copyright and trademark enforcement is many times more common than patent enforcement in the software industry underscores the need for more research in this vein. In this paper, we sought to understand whether different forms of intellectual property are substitutes, as is widely assumed, or if they may also be complements. This question is central to our understanding of how various intellectual property types are interrelated, and how the leveraging of these appropriability mechanisms drives firm performance. We advanced theoretical arguments to make the case that complementarity in IP is plausible, and found supporting evidence for complementarity in the leveraging of copyrights and trademarks by PC software firms. Our results suggest that firms may have simultaneously “discovered” hidden value in both these areas of IP, and thus we highlight an important managerial role in the acquisition and exploitation of intellectual property.

Complementarity in Intellectual Property

Different types of intellectual property have hitherto been typically thought of as substitutes. This view has been bolstered by the inherent tradeoffs between the use of patents and trade secrecy, and the relatively narrow focus on individual inventions as the unit of analysis. We proposed a somewhat broader view of the interactions between different types of IP, where both substitution and complementarity is possible. In our approach, we abstract away from individual inventions to look at the set of multiple inventions contained in a product or a firm. When we shift the level of analysis in this way, we release ourselves from the logic of substituting one type of IP for another – a logic of tradeoffs necessitated when protecting individual inventions. When the analysis is instead focused on a set of inventions, opportunities for IP complementarity arise because firms may extract value by leveraging one type of IP for some inventions while at the same time leveraging another type of IP for other related inventions. Such complementarity may occur due to either production-cost or market-driven interactions, which increase the value of using different types of IP together. This view is consistent with Arora (1995), who analyzed the use of different forms of IP on related inventions at the level of the firm, and

with Cohen et al. (2000), who found that firms appear to use increased levels of both patents and secrecy at the same time.

Our research uncovers robust evidence that the use of two types of intellectual property – copyright and trademark – is strongly correlated in software firms. Our interview evidence also supports the idea that these forms of intellectual property may be leveraged in a complementary fashion. From a managerial perspective, this finding suggests that firms may benefit by paying attention to the complementary configuration of different types of IP to protect their bundles of intellectual assets, rather than focusing on each one in isolation. Both our interview evidence and empirical results corroborate the use of common indivisible inputs in IP strategies, expressly *managerial attention* and *organizational resources*, as an important explanation for our finding of IP complementarity.

Managerial Focus and Attention

Within firms, managerial attention is a scarce and valuable input that invariably faces competing demands. We argued, based both on prior research and our interview findings, that when managerial attention is focused on IP strategy, there likely exist complementary benefits in multiple areas of intellectual property use. In the software industry, far-reaching institutional changes in the early 1990s led to increased attention within software firms to IP strategies, as manifested by increasing levels of patenting. We demonstrated empirically that this focus on IP strategy also spilled over to the firms' leveraging of copyrights and trademarks. Our findings thus shed light on the creation of IP strategy within firms, how managerial attention is related to this domain of activity, and the result of near-simultaneous attention being paid by managers to different types of IP.

Organizational IP Resources

Another hypothesized source of complementarity that we investigated was the existence of organizational resources that could be used to pursue strategies across multiple forms of intellectual property. Research has already described how the development of organizational resources to manage patents enabled semiconductor firms to increase their patenting intensity (Hall and Ziedonis, 2001), and we argue that similar resource deployments are also likely to drive IP complementarity. The firm's

organizational IP resources, which may include attorneys, experienced staff, and internal routines, may provide fungible knowledge, expertise, and efficiencies that can be leveraged across multiple areas of IP, thus allowing the firm to recognize and act on opportunities for rent generation that span these related domains. Our empirical results demonstrate that these organizational resources, as measured by patent attorneys employed in the firm, are significantly correlated with copyright and trademark use, and partially account for the observed complementarity between them. However, the complementarity arising from these organizational resources appears to not be completely distinct from managerial attention to IP. Only when we dropped Microsoft from the sample did we find evidence of a distinct role for this factor, consistent with the idea that in smaller firms IP resources are more likely to be shared across different types of IP.

Limitations and Future Research

While our results are provocative and provide broad support for our hypotheses, our findings are not without limitations. First, the empirical work is set within one segment of the software industry. While software is an important, pervasive, and fast-growing sector of the economy, some of our findings may be inherently industry or segment-specific. It could also be argued that our results may be driven by unobserved heterogeneities among our firms, resulting in a spurious set of empirical findings. Our results are somewhat insulated from this criticism, however, because to be consequential, not only must these heterogeneities be correlated with our variables at the within-firm level and not be accounted for by our controls, but they must also simultaneously explain our finding of IP complementarity (reduce correlation between the unexplained errors). Another limitation is that we do not directly observe the impact of complementary strategies in intellectual property on firm performance. We are therefore drawing inferences from the “revealed preferences” of firms, and must be cautious in making normative prescriptions based on our results. While our choice of these methods is constrained by the available data, future work on IP complementarity may overcome these shortcomings by combining appropriate firm performance data with the use of valid exogenous instruments (Athey and Stern, 1998).

This paper draws attention to the thin body of work on IP strategy in areas other than patents, and offers litigation data as a tool to advancing research in these domains. Our work also suggests potential avenues for research that can enhance our understanding of the strategic interrelationships between different types of IP. While we have shed some light on the drivers of IP substitution and complementarity, future research ought to explore more fully the conditions under which specific types of IP are profitably used as substitutes, or as complements. Another potential direction for research in this area would be to abstract away from the single invention unit of analysis along the temporal dimension, and study the interactions between different types of IP for a related set of inventions over time. Furthermore, our results provide robust evidence that certain common managerial and resource inputs may be in part responsible for complementarity in IP. Because we lack evidence for market-driven complementarity in IP, however, future research on “demand-side” complementarities would be welcome. Finally, our results do not specifically explain why some firms paid greater attention to IP strategy and developed superior organizational IP resources over certain time frames. Future research can help us understand when firms react to changes in their intellectual property environments with responsive strategies, and when they do not.

References

- Arora, A. 1995. Licensing tacit knowledge: Intellectual Property Rights and the Market for Know-how. *Economics of Innovation and New Technology* **4** 41-59.
- Arora, A. 1996. Testing for Complementarities in Reduced-Form Regressions: A Note. *Economics Letters* **50(1)** 51-55.
- Arundel, A. 2001. The relative effectiveness of patents and secrecy for appropriation. *Research Policy* **30** 611-624.
- Arundel, A., I. Kabla. 1998. What percentage of innovations are patented? Empirical estimates for European firms. *Research Policy* **27** 127-141
- Athey, S., S. Stern. 1998. An Empirical Framework for Testing Theories About Complementarity in Organizational Design. NBER Working Paper 6600, Cambridge, MA.
- Berner, R., D. Kiley. 2005. Global Brands: Interbrand-Business Week Survey. *Business Week*, August 1.
- Chandler, A. D., T. Hikino. 1990. *Scale and Scope : The Dynamics of Industrial Capitalism*. Belknap Press, Cambridge, MA.
- Cohen, M.D., J.G. March, J.P. Olsen. 1972. A Garbage Can Model of Organizational Choice. *Administrative Science Quarterly* **17** 1-25.
- Cohen, W.M., R.R. Nelson, J. Walsh. 2000. Protecting Their Intellectual Assets: Appropriability Conditions and Why U.S. Manufacturing Firms Patent (or Not). NBER Working Paper 7552, Cambridge, MA.
- Cyert, R.M., J.G. March. 1963. *A Behavioral Theory of the Firm*. Prentice-Hall, Englewood Cliffs, NJ.
- Department of Commerce (1992). A Report to the Secretary of Commerce. *The Advisory Commission on Patent Law Reform*, Washington, D.C..
- Friedman, D.D., W.M. Landes, R.A. Posner. 1991. Some Economics of Trade Secret Law. *Journal of Economic Perspectives* **5(1)** 61-72.

- Graham, S.J.H., D.C. Mowery. 2003. Intellectual Property Protection in the U.S. Software Industry. W.M. Cohen , S. Merrill, eds. *Patents in the Knowledge-based Economy*. National Academies Press, Washington, DC.
- Graham, S.J.H., D.C. Mowery. 2004. Submarines in Software? Continuations in U.S. Software Patenting in the 1980s and 1990s. *Economics of Innovation and New Technology* **14** 443-56.
- Greene, W.H. 1997. *Econometric Analysis*, Third Edition. Upper Saddle River, NJ: Prentice-Hall.
- Griliches, Z. 1990. Patent Statistics as Economic Indicators: A Survey. *Journal of Economic Literature* **28** 1661-1707.
- Griliches, Z., A. Pakes. 1980. Patents and R&D at the Firm. Level: A First Report. *Economic Letters* **5** 377-381.
- Harabi, N. 1995. Sources of Technical Progress: Evidence from Swiss Industry. *Economics of Innovation and New Technology* **4** 67-76.
- Horstman, I., G. MacDonald, A. Slivinsky. 1985. Patents as information transfer mechanism. *Journal of Political Economy* **93** 837-858.
- Kahn, A. 1962. The Role of Patents. J. Miller, ed. *Competition, Cartels, and their Regulation*. North-Holland, Amsterdam.
- Levin, R.C., A.K. Klevorick, R.R. Nelson, S.G. Winter. 1987. Appropriating the Returns from Industrial Research and Development. *Brookings Papers on Economic Activity* **3** 783-820.
- Liebeskind, J.P. 2000. Property Rights, Secrecy, and the Boundaries of the Firm: Theory and Evidence. USC Marshall School of Business Working Paper, Los Angeles.
- Lippman, S.A. and R.P. Rumelt. 2003. The Bargaining Perspective. *Strategic Management Journal* **24** 1069-86.
- Lybecker, K. 2003. Product Piracy: The Sale of Counterfeit Pharmaceuticals in Developing Countries. Drexel University Working Paper, Philadelphia.
- Machlup, F. 1962. An Economic Review of the Patent System. J. Miller, ed. *Competition, Cartels, and their Regulation*. Amsterdam: North-Holland.

- March, J.G. 1991. How Decisions Happen in Organizations. *Human-Computer Interaction* **6(2)** 95-117
- Milgrom, P., J. Roberts. 1990. The Economics of Modern Manufacturing: Technology, Strategy, and Organization. *American Economic Review* **80(3)** 511-528.
- Milgrom, P., Y. Qian, J. Roberts. 1991. Complementarities, Momentum, and the Evolution of Modern Manufacturing. *American Economic Review* **81** 85-89.
- Occasio, W. 1997. Towards an attention-based view of the firm. *Strategic Management Journal* **18** 187-206.
- Plant, A. 1934a. The Economic Theory Concerning Patents for Inventions. *Economica* **1(1)** 30-51.
- Plant, A. 1934b. The Economic Aspects of Copyright in Books. *Economica* **1(2)** 167-195.
- Peteraf, M. 1993. The Cornerstones of Competitive Advantage: A Resource-based View. *Strategic Management Journal* **14(2)** 179-191.
- Rumelt, R. 1991. How much does industry matter? *Strategic Management Journal* **12** 167-186.
- Samuelson, P. 1984. CONTU Revisited: The Case against Copyright Protection for Computer Programs in Machine-Readable Form. *Duke Law Journal* **September** 663- 752.
- Samuelson, P. and R.J. Glushko. 1990. Survey on the Look and Feel Lawsuits. *Communications of the ACM* **33(5)** 483-95.
- Simon, H.A. 1947. *Administrative Behavior*. The Free Press, New York.
- Somaya, D. 2003. Strategic Determinants of Decisions Not to Settle Patent Litigation. *Strategic Management Journal* **24** 17-38.
- Somaya, D. 2004. Firm Strategies and Trends in Patent Litigation in the United States. G. Libecap, ed. *Advances in the Study of Entrepreneurship, Innovation and Economic Growth*. JAI Press, Greenwich, CT.
- Teece, D.J. 1986. Profiting from Technological Innovation: Implications for Integration, Collaboration, Licensing and Public Policy. *Research Policy* **15(6)** 285-305.
- Weick, K.E. 1979. *The social psychology of organizing*. Addison Wesley, Reading, MA.

Williamson, O.E. 1985. *The Economic Institutions of Capitalism*. The Free Press, New York.

Ziedonis, R. H. 2004. Don't Fence Me In: Fragmented Markets for Technology and the Patent Acquisition Strategies of Firms. *Management Science* **50(6)** 804-820.

Table 1: Interview Responses, IP Managers at Large Packaged Software firms

Manager	Firm	Are different types of IP used as substitutes or complements (Is litigation in one type of IP likely to increase litigation in other types of IP)? What makes it more likely that different types of IP will be used together?
Mgr. 1	A	"For plaintiffs involved in [IP] suits, sensitivity to IP in one area makes one sensitive to IP in other areas. Companies will naturally expand into other areas--so if a young company begins using trademark, it is much easier to begin using patent and copyright." "This expanding into new areas happens in two ways. First, a firm using outside counsel can be sensitized by the outside lawyers, it can sensitize business managers inside the company on [sic] IP issues. Second, companies may hire an in-house counsel with expertise in IP, who then educates the company to recognize the value of IP."
Mgr. 2	B	"We used to look at [patent, copyright, and trademark] as separate, but now we're looking at them together, and increasingly looking at IP as an asset, like our code." "I would naturally expect that if a company is trading in one kind of IP that it will value all types of IP. Our first litigation was copyright litigation [] at a time when patents were not even available on software. Lawyers will use what they have on hand: if a tool is available, the tool will be used by the company." "[At our company], it is true that both the organization and the management has [sic] become more sophisticated in these issues. It has been an evolution at [our company], driven by an increasing sophistication of our IP legal department, a sophistication in these matters by our clients and partners who helped educate us, and also in our business managers."
Mgr. 3	C	"The different IPRs offer protection against different actions. Trade secret protects the stealing of source code, copyright protects against piracy, and patent. It really doesn't protect any important assets of the company. [Our company], though, is being forced by the actions of others to accumulate patents." "Some companies have created profit centers in their legal departments. They see it as an opportunity to monetize their IP. This approach can be a result of legal department theory, or of upper management demanding that the legal department takes that tack."
Mgr. 4	D	"It seems natural that [different IP types] are all used together. Because software can be protected by all three types [of IP], we see them used together. Lawyers throw all the tools in the box at you."
Mgr. 5	E	"It's not surprising to me [that different IP types are used together]--firms involved in one are likely to be involved in others, purely because of the relationship between IP types in the law."
Mgr. 6	A	"I think two things are tied together here: On the one hand, when a company is involved in IP--has an IP strategy--it may lead the company to thinking about these issues holistically. But some may not think about it that way. If you're a big player, you will find yourself in litigation, and because there are a common set of disputes, if you're big enough, you will be forced to be involved in all types of IP disputes. It comes with the territory."

Table 2: Descriptive Statistics of PC Software Firm Sample

Variables	Measures (Firm-Year)	Mean	Standard Deviation			Min	Max
			Overall	Between	Within		
Copyright Use	Suit-Years of Copyright Litigation	1.29	9.13	6.04	6.26	0.00	143.21
Trademark Use	Suit-Years of Trademark Litigation	0.21	1.14	0.75	0.78	0.00	16.37
Managerial Attention (to IP)	Patents Applied for by the Firm	7.98	52.46	45.90	14.96	0.00	589.00
Organizational IP Resources	Number of Patent Attorneys Employed by the Firm	0.17	1.13	0.93	0.49	0.00	14.00
Revenues	Firm Net Sales (1990 \$ Mn, using GDP deflator)	242.4	1067.4	930.6	315.4	0.5	12163.6
Firm Age	Age of Firm in Years	10.36	5.28	5.57	1.43	1.00	30.00
Employees	Number of Firm Employees (in thousands)	0.91	2.74	2.43	0.54	0.01	27.06
Total Assets	Total Firm Assets (1990 \$ Mn, using GDP deflator)	292.2	1467.2	1237.8	557.9	1.0	18775.3
R & D Expense	Firm R&D Expenses (1990 \$ Mn, using GDP deflator)	43.87	181.00	155.58	60.95	0.00	2101.16
Pretax Income	Profits Before Tax (1990 \$ Mn, using GDP deflator)	66.8	484.2	408.8	184.8	-128.0	5976.8
Copyright Registrations	Number of Copyrights Registered by the Firm	6.41	25.34	16.27	14.64	0.00	321.00
Trademark Registrations	Number of Trademarks Registered by the Firm	3.94	9.31	7.75	3.87	0.0	114.0
Total Observations = 290							
Total Firms = 88							

Table 3: Fixed Effects Seemingly Unrelated Regressions (SUR) of IP Use by PC Software Firms

<i>Model Type:</i>	Model(s) 1		Model 2		Model 3		Model 4		Model 5	
	Indiv. Fixed Eff.		Fixed Eff. SUR		Fixed Eff. SUR		Fixed Eff. SUR		Fixed Eff. SUR	
	Copy-right	Trade-mark	Copy-right	Trade-mark	Copy-right	Trade-mark	Copy-right	Trade-mark	Copy-right	Trade-mark
Constant	-1.871 (1.184)	0.149 (0.265)	-1.863 (1.753)	0.03 (0.389)	-0.882 (1.642)	0.16 (0.378)	-0.882 (1.725)	0.22 (0.381)	-0.712 (1.657)	0.228 (0.378)
Revenues (\$ Mn 1990)	-0.023* (0.003)	-0.002* (0.001)	-0.023* (0.004)	-0.003* (0.001)	-0.020* (0.003)	-0.002+ (0.001)	-0.020* (0.004)	-0.002+ (0.001)	-0.020* (0.004)	-0.002+ (0.001)
Firm Age (years)	0.054 (0.095)	0.034 (0.022)	0.015 (0.081)	0.014 (0.018)	-0.028 (0.075)	0.007 (0.018)	-0.015 (0.079)	0.008 (0.018)	-0.031 (0.076)	0.006 (0.018)
Employees (x1000)	1.386* (0.531)	-0.378* (0.124)	1.388+ (0.563)	-0.322+ (0.131)	2.482* (0.565)	-0.139 (0.137)	2.174* (0.594)	-0.162 (0.136)	2.581* (0.580)	-0.105 (0.138)
Total Assets (\$ Mn 1990)	0.018* (0.001)	0.003* (0.000)	0.018* (0.001)	0.003* (0.000)	0.017* (0.001)	0.003* (0.000)	0.016* (0.001)	0.003* (0.000)	0.017* (0.001)	0.003* (0.000)
R & D Expense (\$ Mn 1990)	0.016 (0.010)	0.004 (0.002)	0.016 (0.011)	0.003 (0.002)	-0.02 (0.012)	-0.003 (0.003)	-0.002 (0.012)	-0.001 (0.003)	-0.021 (0.012)	-0.003 (0.003)
Pretax Income (\$ Mn 1990)	0.009* (0.003)	-0.003* (0.001)	0.009+ (0.004)	-0.003* (0.001)	0.006 (0.003)	-0.003* (0.001)	0.008+ (0.003)	-0.003* (0.001)	0.006 (0.003)	-0.003* (0.001)
Zero (Copyright or Trademark) Dummy	0.374 (0.454)	-0.455* (0.069)	0.59 (0.458)	-0.429* (0.070)	0.427 (0.434)	-0.424* (0.069)	0.504 (0.453)	-0.444* (0.069)	0.414 (0.437)	-0.436* (0.069)
Copyright Registrations	0.008 (0.006)		0.007 (0.006)		0.009 (0.005)		0.009 (0.006)		0.010 (0.005)	
Trademark Registrations		-0.016+ (0.007)		(0.003) (0.007)		(0.004) (0.007)		(0.008) (0.007)		(0.007) (0.007)
Managerial Attention (to IP)					0.113* (0.022)	0.019* (0.005)			0.102* (0.027)	0.013+ (0.006)
Organizational IP (Legal) Resources							1.988* (0.594)	0.459* (0.133)	0.533 (0.686)	0.264 (0.162)
Correlation (Rho)	-		0.314*		0.248*		0.246*		0.228*	
Breusch-Pagan Chi-Squared (1)	-		(28.66)		(17.80)		(17.58)		(15.07)	
Chi-Sq (2): Var.=0 in Both Equations	-		-		32.69*		18.59*		2.84	
Variable Name	-		-		Managerial Attention to IP		Organizational IP Legal Resources		Organizational IP Legal Resources	
No. Of Observations	290	290	290	290	290	290	290	290	290	290
Fixed Firm Effects	88	88	88	88	88	88	88	88	88	88
Year Dummies	13	13	13	13	13	13	13	13	13	13

Standard errors in parantheses

+ significant at 5%; * significant at 1%

Table 4: Fixed Effects SUR of IP Use by PC Software Firms (without Microsoft)

Model Type:	Model(s) 1		Model 2		Model 3		Model 4		Model 5	
	Indiv. Fixed Eff.		Fixed Eff. SUR		Fixed Eff. SUR		Fixed Eff. SUR		Fixed Eff. SUR	
	Copy-right	Trade-mark	Copy-right	Trade-mark	Copy-right	Trade-mark	Copy-right	Trade-mark	Copy-right	Trade-mark
Constant	-1.966+ (0.914)	0.197 (0.260)	-3.138+ (1.357)	0.029 (0.382)	-2.313+ (1.027)	0.087 (0.365)	-1.951 (1.112)	0.175 (0.368)	-1.836 (0.956)	0.162 (0.363)
Revenues (\$ Mn 1990)	-0.009* (0.003)	-0.001 (0.001)	-0.009* (0.003)	-0.001 (0.001)	0 (0.003)	0 (0.001)	0.001 (0.003)	0 (0.001)	0.004 (0.003)	0.001 (0.001)
Firm Age (years)	0.234* (0.074)	0.033 (0.021)	0.026 (0.062)	0.01 (0.018)	-0.002 (0.047)	0.006 (0.017)	-0.006 (0.051)	0.006 (0.017)	-0.013 (0.044)	0.004 (0.017)
Employees (x1000)	-0.04 (0.614)	-0.294 (0.177)	-0.039 (0.654)	-0.3 (0.188)	-0.124 (0.493)	-0.315 (0.179)	0.34 (0.533)	-0.243 (0.181)	0.115 (0.460)	-0.274 (0.179)
Total Assets (\$ Mn 1990)	0.003+ (0.001)	0.002* 0.000	0.003+ (0.002)	0.002* 0.000	0 (0.001)	0.002* 0.000	-0.002 (0.001)	0.001* 0.000	-0.002 (0.001)	0.001* 0.000
R & D Expense (\$ Mn 1990)	0.030* (0.009)	0.003 (0.003)	0.030* (0.009)	0.003 (0.003)	-0.006 (0.008)	-0.002 (0.003)	0.007 (0.008)	-0.001 (0.003)	-0.01 (0.007)	-0.003 (0.003)
Pretax Income (\$ Mn 1990)	0.005 (0.003)	-0.005* (0.001)	0.005 (0.003)	-0.005* (0.001)	0.004 (0.002)	-0.005* (0.001)	0.005+ (0.003)	-0.005* (0.001)	0.005+ (0.002)	-0.005* (0.001)
Zero (Copyright or Trademark) Dummy	-0.3 (0.349)	-0.480* (0.067)	-0.089 (0.359)	-0.456* (0.069)	-0.435 (0.280)	-0.447* (0.068)	-0.38 (0.301)	-0.472* (0.068)	-0.525+ (0.260)	-0.461* (0.068)
Copyright Registrations	(0.002) (0.004)		(0.001) (0.004)		0.002 (0.003)		0.002 (0.004)		0.003 (0.003)	
Trademark Registrations		(0.016) (0.008)		(0.010) (0.009)		(0.005) (0.009)		(0.010) (0.008)		(0.006) (0.009)
Managerial Attention (to IP)					0.161* (0.015)	0.022* (0.006)			0.119* (0.016)	0.015+ (0.006)
Organizational IP (Legal) Resources							3.668* (0.406)	0.531* (0.137)	2.096* (0.407)	0.332+ (0.158)
Correlation (Rho)	-		0.258*		0.082		0.110*		0.016*	
Breusch-Pagan Chi-Squared (1)	-		(18.98)		(1.90)		(3.46)		(0.08)	
Chi-Sq (2): Var.=0 in Both Equations	-		-		131.1*		90.26*		30.60*	
Variable Name	-		-		Managerial Attention to IP		Organizational IP Legal Resources		Organizational IP Legal Resources	
No. Of Observations	286	286	286	286	286	286	286	286	286	286
Fixed Firm Effects	87	87	87	87	87	87	87	87	87	87
Year Dummies	13	13	13	13	13	13	13	13	13	13

Standard errors in parantheses

+ significant at 5%; * significant at 1%

Figure 1: Patent Litigation and Patenting by PC Software (Softletter 100) Firms

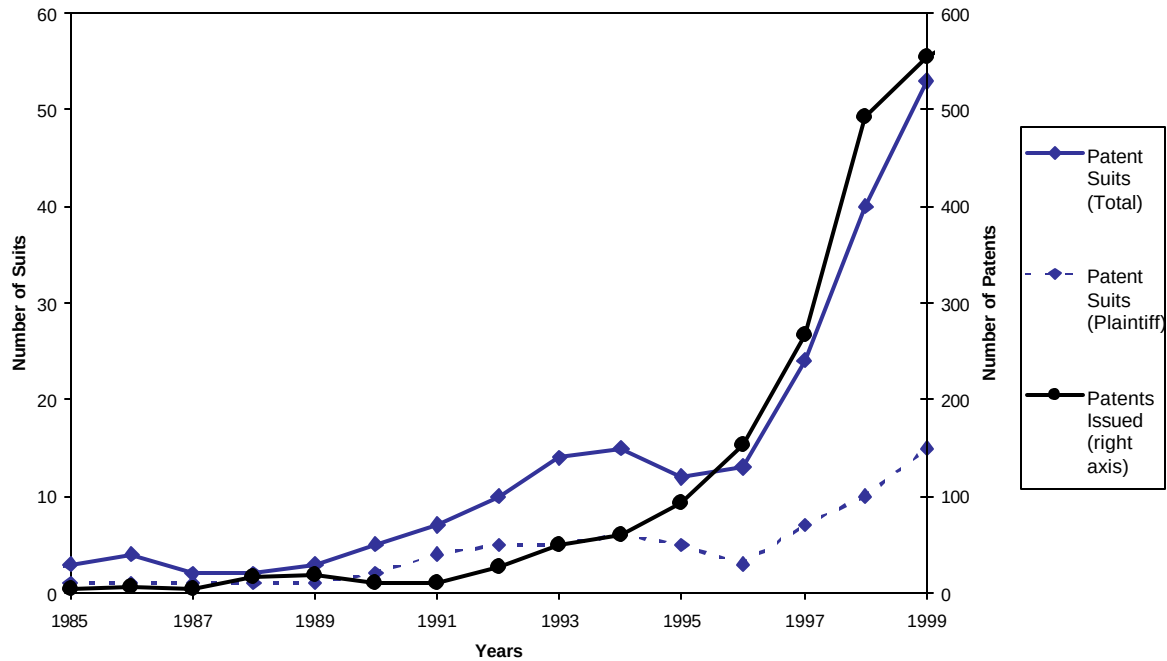


Figure 2: Copyright, and Trademark Litigation of Softletter 100 Firms (Total and as Plaintiffs)

